NATIONAL MATHEMATICS CONTEST
A-LEVEL PAPER.

SATURDAY 28TH MAY, 2022 9:00 AM - 12:15 PM

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

(a) *This competition is conducted on the assumption that proper security is maintained. UMS reserves the right, should there be evidence or suspicion of any malpractice, to reject scripts from a particular school/institution.*

(b) *There are* **two** *sections and each carries* 50 *marks. Attempt all questions in exactly three hours fifteen minutes (*$9:00am$ *to* $12:15pm$*).*

(c) *Indicate your names, birth date, gender, class, school and district* ***on all*** *your answer sheets.*

(d) *Marks will be awarded for only answers for which a clear and logical layout of the working is provided. A correct solution, poorly presented, will not earn full marks.*

(e) *Answer scripts of only registered students should be returned to the contest coordinator, latest Monday 30th May, 2022.*

(f) *ALL participants who score a pass mark are cordially invited to the certificate and prize giving ceremony on Saturday* 23*rd July,* 2022 *at 8:00am at Makerere University, Kampala.*

SECTION A : 5 marks each

1. What is the value of the following product?

$$2^{2022}\cos\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{8}\right)\cos\left(\frac{\pi}{16}\right)\cos\left(\frac{\pi}{32}\right)\cdots\cos\left(\frac{\pi}{2^{2022}}\right)$$

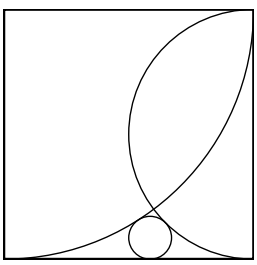
2. Find the function $f(x)$ such that $f(x)+f\left(\dfrac{1}{1-x}\right)=x$

3. Let $a_1=1, a_2=1+\dfrac{1}{1}, a_3=1+\dfrac{1}{1+\dfrac{1}{1}}, a_4=1+\dfrac{1}{1+\dfrac{1}{1+\dfrac{1}{1}}}$ and continued in the same manner.

   Find a_{10}.

4. Suppose $x\neq y$ and the first terms of the two arithmetic sequences are x, a_1, a_2, a_3, y, a_4 and b_1, x, b_2, b_3, y, b_4. What is the value of $\dfrac{b_4-b_3}{a_2-a_1}$?

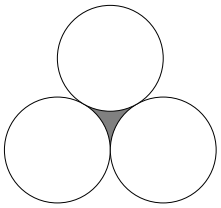
5. In a square with side length 1, a quarter circle is drawn with center at the top left corner and passing through the top right and bottom left corners, and a half circle is drawn with the right side of the square as its diameter. A smaller circle is drawn tangent to the quarter circle, the half circle, and the bottom edge (as shown). Determine its radius.



6. Given $x+\dfrac{1}{x}=-1$. Determine $x^{88}+\dfrac{1}{x^{88}}$.

7. Assume that real numbers a and b satisfy $ab+\sqrt{ab+1}+\sqrt{a^2+b}\cdot\sqrt{b^2+a}=0$. Find, with proof, the value of $a\sqrt{b^2+a}+b\sqrt{a^2+b}$.

8. Given three circles of radius r, tangent to each other as shown in the following diagram, what is the area for the shaded region?



9. Darth Vader urgently needed a new Death Star battle station. He sent requests to four planets asking how much time they would need to build it. The Mandalorians answered that they can build it in one year, the Sorganians in one and a half year, the Nevarroins in two years, and the Klatooinians in three years. To expedise the work Darth Vader decided to hire all of them to work together. The Rebels need to know when the Death Star is operational. Can you help the Rebels to find the number of days needed if all four planets work together? We assume that one year=365 days.

10. The numbers 1447, 1005 and 1231 have something in common: each is a 4-digit number beginning with 1 that has exactly two identical digits. How many such numbers are there?

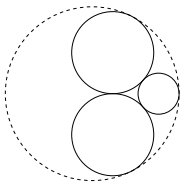
SECTION B : 10 marks each

11. Find all the real values that satisfy
$$\frac{27^x+343^x}{63^x+147^x}=\frac{37}{21}$$

12. (a) Find the product $a\cdot b$, given that
$$a = \sqrt{6+\sqrt{6+\sqrt{6+\sqrt{\cdots}}}}$$
$$b = \sqrt{9-\sqrt{9-\sqrt{9-\sqrt{\cdots}}}}$$

(b) Find all sequences $a_1, a_2, a_3, \ldots$ of real numbers such that for all positive integers $m, n \geq 1$, we have
$$\begin{aligned} a_{m+n} &= a_m + a_n - mn \\ a_{mn} &= m^2 a_n + n^2 a_m + 2a_m a_n. \end{aligned}$$

13. Two circles of radius 1 and one circle of radius 1/2 are drawn on a plane so that each of them is touching the other two at one point as shown. What is the radius of the largest circle (dashed) tangent to all three of these circles?



14. (a) Let a, b, c, and d be integers satisfying $a\log_{10}2+b\log_{10}3+c\log_{10}5+d\log_{10}7=2022$. What is $a+b+c+d$?

(b) Consider the sequence $\{a_n\}$ given by $a_{n+1}=\dfrac{2(n+2)}{(n+1)}a_n$, with $a_1=2$. Compute the value of
$$\frac{a_{2022}}{a_1+a_2+\cdots+a_{2021}}.$$

15. Given a triangle A, B, C, X is on side AB, Y is on side AC, and P and Q are on side BC such that $AX=AY, BX=BP$ and $CY=CQ$. Let XP and YQ intersect at T. Prove that AT passes through the midpoint of PQ.

1. **Solution :** *The key is to repeatedly use the equality:*

$$\cos x = \frac{\sin 2x}{2\sin x}$$

The product then equals:

$$\begin{aligned} & 2^{2022}\frac{\sin\left(\frac{\pi}{2}\right)}{2\sin\left(\frac{\pi}{4}\right)}\cdot\frac{\sin\left(\frac{\pi}{4}\right)}{2\sin\left(\frac{\pi}{8}\right)}\cdots\frac{\sin\left(\frac{\pi}{2^{2021}}\right)}{2\sin\left(\frac{\pi}{2^{2022}}\right)} \\ = & \frac{2^{2022}}{2^{2021}}\frac{\sin\left(\frac{\pi}{2}\right)}{\sin\left(\frac{\pi}{2^{2022}}\right)} \\ = & 2\frac{1}{\sin\left(\frac{\pi}{2^{2022}}\right)} \\ = & 2\csc\frac{\pi}{2^{2022}} \end{aligned}$$

■

2. **Solution :** *Replace x with* $\dfrac{1}{1-x}$

$$f(x)+f\left(\frac{1}{1-x}\right) = x \tag{1.1}$$

$$f\left(\frac{}{1-x}\right)+f\left(\frac{1}{1-\frac{1}{1-x}}\right) = \frac{1}{1-x}$$

But $\quad \dfrac{1}{1-\frac{1}{1-x}} = \dfrac{1}{\frac{(1-x)-1}{1-x}} = \dfrac{1}{\frac{-x}{1-x}} = \dfrac{1-x}{-x} = \dfrac{-1}{x}$

$$f\left(\frac{1}{1-x}\right)+f\left(\frac{x-1}{x}\right) = \frac{1}{1-x} \tag{1.2}$$

Replace x with $\dfrac{x-1}{x}$

$$\begin{aligned} f\left(\frac{x-1}{x}\right)+f\left(\frac{1}{1-\frac{x-1}{x}}\right) &= \frac{x-1}{x} \\ f\left(\frac{x-1}{x}\right)+f\left(\frac{1}{1-\frac{x-1}{x}}\right) &= \frac{x-1}{x} \\ f\left(\frac{x-1}{x}\right)+f\left(\frac{x}{1}\right) &= \frac{x-1}{x} \\ f\left(\frac{x-1}{x}\right)+f(x) &= \frac{x-1}{x} \end{aligned} \tag{1.3}$$

(1.1)+(1.3)−(1.2) we have,

$$\begin{aligned}
2f(x) &= x + \left(\frac{x-1}{x}\right) - \frac{1}{(1-x)} \\
2f(x) &= x + \frac{x-1}{x} + \frac{1}{x+1} \\
2f(x) &= \frac{x(x)(x-1) + (x-1)(x-1) + x}{x(x-1)} \\
2f(x) &= \frac{x^2(x-1) + (x-1)(x-1) + x}{x(x-1)} \\
2f(x) &= \frac{x^3 - x^2 + x^2 - 2x + 1 + x}{x(x-1)} \\
2f(x) &= \frac{x^3 - x + 1}{x(x-1)} \\
f(x) &= \frac{x^3 - x + 1}{2x(x-1)}
\end{aligned}$$

■

3. **Solution :** $\frac{89}{55}$.

We note that $a_1 = \frac{1}{1}, a_2 = \frac{2}{1}, a_3 = \frac{3}{2}, a_4 = \frac{5}{3}$ *and, in general, if* $a_n = \frac{p}{q}$ *then* $a_{n+1} = \frac{p+q}{p}$ *. We can prove this via the identity* $a_{n+1} = 1 + \frac{1}{a_n}$.

In any event, we observe that the numerator and denominator are Fibbonacci numbers and just carry out this pattern to a_{10}*:*

$$a_5 = \frac{8}{5}, a_6 = \frac{13}{8}, a_7 = \frac{21}{13}, a_8 = \frac{34}{21}, a_9 = \frac{55}{34}, a_{10} = \frac{89}{55}$$

■

4. **Solution :** *Let* d_1 *be the difference of the first arithmetic sequence, and let* d_2 *be the difference of the second arithmetic sequence. Considering the first sequence we get* $y - x = 4d_1$*, and considering the second sequence we obtain* $y - x = 3d_2$*. In that case,* $\frac{b_4 - b_3}{a_2 - a_1} = \frac{2d_2}{d_1} = \frac{8}{3}$. ■

5. **Solution :** *Assume the diagram is oriented so that the bottom left has coordinates* $(0,0)$ *and the top right* $(1,1)$*. Write* (a,x) *for the coordinates of the smallest circle; then the radii of the three circles are (from smallest to largest)* $x, \frac{1}{2}$ *, and* 1*.*
The line from $(0,1)$ *to* (a,x) *has length* $1+x$*, as it comprises two radii. Therefore we have* $a^2 + (1-x)^2 = (1+x)^2$*, and hence* $a^2 = 4x$*. Similarly, considering a line from* $\left(1, \frac{1}{2}\right)$ *to* (a,x) *we conclude that* $(1-a)^2 + \left(\frac{1}{2} - x\right)^2 = \left(\frac{1}{2} + x\right)^2$*, which reduces to* $(1-a)^2 = 2x$*.*
So $4x = 2(1-a)^2 = a^2$*, so that* $a^2 - 4a + 2 = 0$*. This quadratic equation has the solutions* $a = 2 \pm \sqrt{2}$*, of which only* $2 - \sqrt{2}$ *makes sense geometrically. Therefore*

$$x = \frac{3}{2} - \sqrt{2}$$

■

6. **Solution :**

Method I

Multiplying by x we obtain,

$$\begin{aligned} x^2+1 &= x \\ x^2+x+1 &= 0 \\ x &= \frac{1\pm\sqrt{1-4(1)(1)}}{2} \\ x &= \frac{1\pm\sqrt{3}i}{2} \end{aligned}$$

Since $x^2+x+1=0$

$$\begin{aligned} a^3-b^3 &= (a-b)(a^2+ab+b^2) \\ x^3-1^3 &= (x-1)(x^2+x+1) \\ x^3-1^3 &= (x-1)(0) \\ x^3-1 &= 0 \\ \Rightarrow\quad x^3 &= 1 \end{aligned}$$

Therefore

$$\begin{aligned} x^{88} &= x\cdot x^{87} \\ 87 &= x^{3\cdot 29} = (x^3)^{29}\ \ 1^{29} = 1 \\ \Rightarrow\quad x^{88} &= x\cdot x^{87} = x\cdot x^{3(29)} = x\cdot(x^3)^{29} \\ &\Leftrightarrow \\ x^{88}+\frac{1}{x^{88}} &= x\cdot(x^3)^{29}+\frac{1}{x\cdot(x^3)^{29}} \\ &= x\cdot(1)^{29}+\frac{1}{x\cdot(1)^{29}} \\ &= x+\frac{1}{x} = -1 \end{aligned}$$

Alternatively;

Method II

One an apply the Demorve's theorem where;

$$\begin{aligned} x^n &= r^n\big(\cos n\theta + i\sin n\theta\big) \\ x &= \frac{1}{2}\pm\frac{\sqrt{3}i}{2} \\ \left(\frac{1}{2}\pm\frac{\sqrt{3}i}{2}\right)^{88} &= r^{88}\big(\cos 88\theta \pm \sin 88\theta\big) \\ r &= \sqrt{\left(\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2} = 1 \end{aligned}$$

$$\begin{aligned}
\theta &= tan^{-1}\left(\frac{y}{x}\right) \quad = \tan^{-1}\left(\frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}}\right) \quad = \tan^{-1}\left(\sqrt{3}\right) \quad = \frac{\pi}{3} \\
\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)^{88} &= (1)^{88}\left(\cos\left(88\cdot\left(\frac{\pi}{3}\right)\right)+i\sin\left(88\cdot\left(\frac{\pi}{3}\right)\right)\right) \\
x^{88}+\frac{1}{x^{88}} &= \left(\cos\left(88\cdot\left(\frac{\pi}{3}\right)\right)+i\sin\left(88\cdot\left(\frac{\pi}{3}\right)\right)\right)+\frac{1}{\left(\cos\left(88\cdot\left(\frac{\pi}{3}\right)\right)+i\sin\left(88\cdot\left(\frac{\pi}{3}\right)\right)\right)}
\end{aligned}$$

and

$$\begin{aligned}
\theta &= tan^{-1}\left(\frac{y}{x}\right) \quad = \tan^{-1}\left(\frac{-\sqrt{3}2}{\frac{1}{2}}\right) \quad = \tan^{-1}\left(-\sqrt{3}\right) \quad = -\frac{\pi}{3} \\
\left(\frac{1}{2}-\frac{\sqrt{3}i}{2}\right)^{88} &= (1)^{88}\left(\cos\left(88\cdot\left(\frac{\pi}{3}\right)\right)-i\sin\left(88\cdot\left(\frac{\pi}{3}\right)\right)\right) \\
&= \left(\cos\left(88\cdot\left(\frac{\pi}{3}\right)\right)-i\sin\left(88\cdot\left(\frac{\pi}{3}\right)\right)\right) \\
x^{88}+\frac{1}{x^{88}} &= \left(\cos\left(88\cdot\left(\frac{\pi}{3}\right)\right)-i\sin\left(88\cdot\left(\frac{\pi}{3}\right)\right)\right)+\frac{1}{\left(\cos\left(88\cdot\left(\frac{\pi}{3}\right)\right)-i\sin\left(88\cdot\left(\frac{\pi}{3}\right)\right)\right)}
\end{aligned}$$

Then one can use the above $x's$ to obtain $x^{88}+\frac{1}{x^{88}}$ ■

7. **Solution :** *Let us rewrite the given equation as follows.*

$$ab+\sqrt{a^2+b}\sqrt{b^2+a}=-\sqrt{ab+1}.$$

Squaring this gives us

$$\begin{aligned}
a^2b^2+2ab\sqrt{a^2+b}\sqrt{b^2+a}+(a^2+b)(b^2+a) &= ab+1 \\
(a^2b^2+a^3)+2ab\sqrt{a^2+b}\sqrt{b^2+a}+(a^2b^2+b^3) &= 1 \\
\left(a\sqrt{b^2+a}+b\sqrt{a^2+b}\right)^2 &= 1 \\
a\sqrt{b^2+a}+b\sqrt{a^2+b} &= \pm 1.
\end{aligned}$$

Next, we show that $a\sqrt{b^2+a}+b\sqrt{a^2+b}>0$. Note that

$$ab=-\sqrt{ab+1}-\sqrt{a^2+b}\cdot\sqrt{b^2+a}<0,$$

so a and b have opposite signs. Without loss of generality, we may assume $a>0>b$. Then rewrite

$$a\sqrt{b^2+a}+b\sqrt{a^2+b}=a\left(\sqrt{b^2+a}+b\right)-b\left(a-\sqrt{a^2+b}\right)$$

and, since $\sqrt{b^2+a}+b$ and $a-\sqrt{a^2+b}$ are both positive, the expression above is positive. Therefore,

$$a\sqrt{b^2+a}+b\sqrt{a^2+b}=1,$$

and the proof is finished. ■

8. **Solution :** *If we connect the centers of each of the circles, we obtain an equilateral triangle, each side of which has a length of* $2r$. *This triangle has a height of* $r\sqrt{3}$, *and thus the area of this triangle is* $\dfrac{(r\sqrt{3})(2r)}{2} = \sqrt{3}r^2$. *To find the shaded area, we take the area of the triangle, and we subtract three copies of the area of the 3 sectors of the circles with central angles*

$$\frac{\pi}{3} \to A = \frac{1}{2}r^2\left(\frac{\pi}{3}\right) = \frac{\pi r^2}{6}.$$

Thus the shaded area is

$$\sqrt{3}r^2 - 3\left(\frac{\pi r^2}{6}\right) = \sqrt{3}r^2 - \left(\frac{\pi r^2}{2}\right) = \left(\sqrt{3} - \frac{\pi}{2}\right)r^2$$

■

9. **Solution :** *Denote by* x *the total amount of work. In one day the Mandalorians complete* $\dfrac{x}{365}$, *the Sorganians* $\dfrac{2}{3}\cdot\dfrac{x}{365}$, *the Nevarroins* $\dfrac{1}{2}\cdot\dfrac{x}{365}$, *and the Klatooinians* $\dfrac{1}{3}\cdot\dfrac{x}{365}$. *Thus, together in one day they complete*

$$\begin{aligned}\frac{x}{365} + \frac{2}{3}\cdot\frac{x}{365} + \frac{1}{2}\cdot\frac{x}{365} + \frac{1}{3}\cdot\frac{x}{365} &= \frac{x}{365}\cdot\left(1 + \frac{2}{3} + \frac{1}{2} + \frac{1}{3}\right)\\ &= \frac{5}{2}\cdot\frac{x}{365} = \frac{x}{146}.\end{aligned}$$

Therefore, working together they will build the battle station in 146 *days.* ■

10. **Solution :** ***Solution 1***

Suppose that the two identical digits are both 1. *Since the thousands digit must be* 1, *only one of the other three digits can be* 1. *This means the possible forms for the number are* $11xy$, $1x1y$, $1xy1$

Because the number must have exactly two identical digits, $x \neq y$, $x \neq 1$, *and* $y \neq 1$. *Hence, there are* $3 \cdot 9 \cdot 8 = 216$ *numbers of this form.*

Now suppose that the two identical digits are not 1. *Reasoning similarly to before, we have the following possibilities:* $1xxy$, $1xyx$, $1yxx$.

Again, $x \neq y$, $x \neq 1$, *and* $y \neq 1$. *There are* $3 \cdot 9 \cdot 8 = 216$ *numbers of this form.*

Thus the answer is $216 + 216 = \boxed{432}$.

Solution 2

Consider a sequence of 4 *digits instead of a* 4*-digit number. Only looking at the sequences which have one digit repeated twice, we notice that the probability that the sequence starts with 1 is* $\frac{1}{10}$. *This means we can find all possible sequences with one digit repeated twice, and then divide by* 10.

If we let the three distinct digits of the sequence be a, b, *and* c, *with* a *repeated twice, we*

can make a table with all possible sequences:

aabc	*abac*	*abca*
baac	*baca*	
bcaa		

There are 6 *possible sequences.*

Next, we can see how many ways we can pick a, b, and c. This is $10(9)(8) = 720$*, because there are* 10 *digits, from which we need to choose* 3 *with regard to order. This means there are* $720(6) = 4320$ *sequences of length* 4 *with one digit repeated. We divide by 10 to get* $\boxed{432}$ *as our answer.*

Solution 3 (Complementary Counting)

We'll use complementary counting. We will split up into 3 *cases: (1) no number is repeated, (2)* 2 *numbers are repeated, and* 2 *other numbers are repeated, (3)* 3 *numbers are repeated, or (4)* 4 *numbers are repeated.*

Case 1: There are 9 *choices for the hundreds digit (it cannot be* 1*),* 8 *choices for the tens digit (it cannot be* 1 *or what was chosen for the hundreds digit), and* 7 *for the units digit. This is a total of* $9 \cdot 8 \cdot 7 = 504$ *numbers.*

Case 2: One of the three numbers must be 1*, and the other two numbers must be the same number, but cannot be* 1 *(That will be dealt with in case 4). There are* 3 *choices to put the* 1*, and there are* 9 *choices (any number except for* 1*) to pick the other number that is repeated, so a total of* $3 \cdot 9 = 27$ *numbers.*

Case 3: We will split it into 2 *subcases: one where* 1 *is repeated* 3 *times, and one where another number is repeated* 3 *times. When* 1 *is repeated* 3 *times, then one of the digits is not* 1*. There are* 9 *choices for that number, and* 3 *choices for its location,so a total of* $9 \cdot 3 = 27$ *numbers. When a number other than* 1 *is repeated* 3 *times, then there are* 9 *choices for the number, and you don't have any choices on where to put that number. So in Case 3 there are* $27 + 9 = 36$ *numbers*

Case 4: There is only 1 *number:* 1111*.*

There are a total of 1000 4*-digit numbers that begin with* 1 *(from* 1000 *to* 1999*), so by complementary counting you get* $1000 - (504 + 27 + 36 + 1) = \boxed{432}$ *numbers.* ■

11. **Solution :**

$$\begin{aligned}
\frac{(3^3)^x + (7^3)^x}{(3^2\cdot 7)^x + (3\cdot 7^2)^x} &= \frac{37}{21}\\
\frac{(3^3)^x + (7^3)^x}{(3\cdot 7)^x\left[3^x + 7^x\right]} &= \frac{37}{21}\\
a^3 + b^3 &= (a+b)(a^2 - ab + b^2)\\
\frac{(3^x)^3 + (7^x)^3}{(3\cdot 7)^x\left[3^x + 7^x\right]} &= \frac{(3^x + 7^x)\left((3^x)^2 - (3^x)(7^x) + (7^x)^2\right)}{(3\cdot 7)^x\left[3^x + 7^x\right]}\\
&= \frac{\cancel{(3^x + 7^x)}\left((3^x)^2 - (3^x)(7^x) + (7^x)^2\right)}{(3\cdot 7)^x\cancel{[3^x + 7^x]}}\\
&= \frac{3^{2x} - 3^x 7^x + 7^{2x}}{(3\cdot 7)^x}\\
\frac{3^{2x} - 3^x 7^x + 7^{2x}}{(3\cdot 7)^x} &= \frac{37}{21}\\
\frac{3^x\cdot 3^x}{3^x\cdot 7^x} - \frac{3^x 7^x}{3^x\cdot 7^x} + \frac{7^x\cdot 7^x}{3^x\cdot 7^x} &= \frac{37}{21}\\
\frac{3^x}{7^x} - 1 + \frac{7^x}{3^x} &= \frac{37}{21}\\
\left(\frac{3}{7}\right)^x - 1\left(\frac{7}{3}\right)^x &= \frac{37}{21}
\end{aligned}$$

Let $u = \left(\frac{3}{7}\right)^x \quad \Rightarrow \quad \left(\frac{3}{7}\right)^x = \frac{1}{\left(\frac{3}{7}\right)^x} = \frac{1}{u}$

$$\begin{aligned}
u - 1 + \frac{1}{u} &= \frac{37}{21}\\
u^2 - u + 1 &= \frac{37}{21}u\\
21u^2 - 21u + 21 &= 37u\\
21u^2 - 58u + 21 &= 0
\end{aligned}$$

Solving the quadratic equation

$$\begin{aligned}
21u^2 - 58u + 21 &= 0\\
21u^2 - 9u - 49u + 21 &= 0\\
3u(7u-3) - 7(7u-3) &= 0\\
(7u-3)(3u-7) &= 0\\
u = \frac{3}{7} \quad &or \quad u = \frac{7}{3}\\
\left(\frac{3}{7}\right)^x = \left(\frac{3}{7}\right) \quad &or \quad \left(\frac{3}{7}\right)^x = \left(\frac{7}{3}\right)\\
\left(\frac{3}{7}\right)^x = \left(\frac{3}{7}\right)^1 \quad &or \quad \left(\frac{3}{7}\right)^x = \left(\frac{3}{7}\right)^{-1}\\
x = 1 \quad &or \quad x = -1
\end{aligned}$$

■

12. (a) **Solution :** *Notice that $a^2 = 6 + a$; therefore, $a = 3$ (we cannot have $a = -2$). Similarly, note that $b^2 = 9 - b$; thus, $b = \dfrac{(\sqrt{37} - 1)}{2}$ (we keep only the positive value). From these, it is simple to compute their product as $\dfrac{3}{2}(\sqrt{37} - 1)$.* ■

(b) **Solution :** *Multiplying the second equation by 2 and adding m^2n^2 to both sides, we find*

$$2a_{mn} + (mn)^2 = (2a_m + m^2)(2a_n + n^2).$$

Therefore, define $b_n = 2a_n + n^2$, so the second equation can be written as $b_{mn} = b_m b_n$. Multiplying the first equation by 2 and adding $(m + n)^2$, we find

$$2a_{m+n} + (m + n)^2 = (2a_m + m^2) + (2a_n + n^2).$$

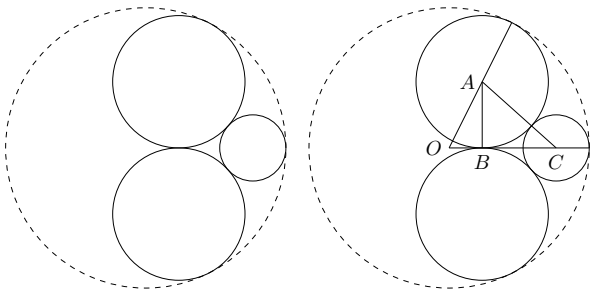
Therefore, $b_{m+n} = b_m + b_n$. Therefore, it suffices to find sequences $b_1, b_2, b_3, \ldots$ that satisfy both $b_{mn} = b_m b_n$ and $b_{m+n} = b_m + b_n$.

We claim that if $b_{m+n} = b_m + b_n$ for all positive integers m, n, then $b_n = nb_1$. Certainly this is true for $n = 1$. If $b_k = kb_1$, then $b_{k+1} = b_k + b_1 = kb_1 + b_1 = (k + 1)b_1$, so by induction, it follows that $b_n = nb_1$ for all positive integers n. Additionally, we observe that any sequence of this form will satisfy $b_{m+n} = b_m + b_n$.

Substituting $m = n = 1$ into $b_{mn} = b_m b_n$, we find $b_1 = b_1^2$, so $b_1 = 0$ or $b_1 = 1$. Combining this with the previous paragraph, the only possible sequences are $b_n = 0$ or $b_n = n$. Indeed, both sequences satisfy both $b_{m+n} = b_m + b_n$ and $b_{mn} = b_m b_n$.

Therefore, since $a_n = \dfrac{b_n - n^2}{2}$, it follows that the two possible sequences are $a_n = -\dfrac{n^2}{2}$ or $a_n = \dfrac{n - n^2}{2}$. ■

13. **Solution :** *In the diagram, label points O (the center of the largest circle), A (the center of one circle of radius 1), B (the intersection of the two circles of radius 1), and C (the center of the circle of radius $1/2$).*



The points O and A are collinear with the point of tangency of the circle of radius 1 with the largest circle, and the radius can then be expressed as $OA + 1$.

The points O, B and C are collinear with the point of tangency of the circle of radius $1/2$ with the largest circle. The radius can also be expressed as $OB + BC + 1/2$.

Note that $AC = 1 + 1/2 = 3/2$; by the Pythagorean Theorem in triangle $\triangle ABC$, we have $BC = \sqrt{5}/2$.

This gives two equations with two variables:

$$\begin{aligned} OA+1 &= OB+\frac{1+\sqrt{5}}{2} && \text{the two radii have the same length} \\ OA^2 &= OB^2+1 && \text{Pythagorean Theorem on triangle } \triangle OAB \end{aligned}$$

Solving for OA, for instance, gives us that $r = OA+1 = 1+\frac{\sqrt{5}}{2}$. ■

14. (a) **Solution :** *Raising both sides to the tenth power, we get*

$$2^a 3^b 5^c 7^d = 10^{2022},$$

so $a = c = 2022$ and $b = d = 0$. So the answer is 4044. ■

(b) **Solution :** *From the definition, we have*

$$\begin{aligned} a_n &= \frac{2(n+1)}{n}a_{n-1} \\ &= \frac{2(n+1)}{n}\cdot\frac{2n}{n-1}a_{n-2} \\ &= \cdots \\ &= \frac{2(n+1)}{n}\cdot\frac{2n}{n-1}\cdots\frac{2\cdot 3}{2}a_1 \\ &= 2^{n-1}(n+1) \end{aligned}$$

The sum of the first n terms of $\{a_n\}$, denoted by S_n, is given by:

$$S_n = 2+2\times 3+2^2\times 4+\cdots+2^{n-1}(n+1).$$

Then

$$2S_n = 2^2+2^2\times 3+2^3\times 4+\cdots+2^n(n+1),$$

so the difference of the above two equations gives

$$\begin{aligned} S_n &= 2^n(n+1)-(2+2+2^2+\cdots+2^{n-2}+2^{n-1}) \\ &= 2^n(n+1)-2^n \\ &= 2^n n. \end{aligned}$$

Therefore,

$$\frac{a_{2022}}{a_1+a_2+\cdots+a_{2021}} = \frac{2^{2021}\times 2023}{2^{2021}\times 2021} = \frac{2023}{2021}.$$

■

15. **Solution :**

Let AT intersect BC at M. Let R, S be on sides AB, AC, respectively such that $MR||XP$ and $MS||YQ$. Then since $BX = BP, BR = BM$. Therefore, $XR = PT$. Similarly, $YS = QT$.

We need to show that $PT = QT$. It suffices to show that $XR = YS$.

Since $RM||XT$ and $SM||YT$, $\frac{AR}{AX} = \frac{AM}{AT} = \frac{AS}{AY}$. Since $AX = AY, AR = AS$. Therefore, $RX = SY$, as desired.

■